MIL-C-82283B(NU)
16 March 1984
SUPERSEDING
MIL-C-82283A(SA)
13 December 1974

MILITARY SPECIFICATION

CLOTH, TWILL, COTTON LIGHTWEIGHT

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a lightweight cotton twill cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton
PPP-P-1134 - Packaging, Packing of Cotton and Cotton-Synthetic Fiber-Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580).

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 4.3 and 6.2), the contractor shall furnish a sample for first article inspection and approval.

3.3 Material. (see 6.5).

3.3.1 Yarn. The yarn shall be made of cotton which has been cleaned, carded, drawn, and spun into singles for both the warp and the filling when tested as specified in 4.4.1 and 4.5.

3.4 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical Requirements

Weight oz per sq yd (g/m2)		Yarns per inch (minimum)(cm)		Breaking Strength lbs (min) (Newton)	
Min.	Max.	Warp	Filling	Warp	Filling
5.5 (186.5)	6.0 (203.4)	72 (28.3)	58 (22.8)	50 (222.5)	60 (267)

3.4.1 Weave. The weave shall be a $\frac{2}{1}$ right or left hand twill.

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* 3.4.2 Width. The width of the cloth shall be as specified (see 6.2), and shall be the minimum width, inclusive of selvage. When woven on shuttless looms the width shall be exclusive of selvages.

3.5 Color. Unless otherwise specified (see 6.2), the finished cloth shall be dyed to match Olive Green 107.

3.5.1 Sulfur content. The use of sulfure dyes or dyes containing elementary sulfur sulfur compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When a standard sample is not available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

3.5.2 Matching. The color of the finished cloth shall match the standard shade under artificial daylight, having a color temperature of 7500 (+ 500) Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2850 (+ 100) Kelvin.

3.5.3 Colorfastness. The finished cloth shall show fastness to light, crocking, perspiration, laundering (after 3 cycles) equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available, the cloth shall show a minimum of "good" fastness to light, perspiration, laundering (after 3 cycles) and a Munsell value for crocking not lower than 8.5. Tests shall be performed as specified in 4.5.

3.6 Finish. The cloth shall be closely singed, fully mercerized, and have a clear lustrous finish when tested as specified in 4.4.1.

3.6.1 Nonfibrous material. The finished cloth shall not exceed 2.0 percent starch and protein content including chloroform-soluble and wate-soluble material when tested as specified in 4.5.

3.7 Dimensional stability. The finished cloth shall not shrink more than 2.0 percent in either the direction of the warp or the filling when tested as specified in 4.5. The pre-shrinking process used shall not be identified by name or trademark, either on the cloth, ticket or package.

3.8 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.5.

3.10 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.4m). Each length shall be put up on full rolls as specified in PPP-P-1134.

3.11 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.12 Face identification. The face side of the cloth shall be identified by stamping the word "FACE" on each end of the roll or piece.

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3.13 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$), when calculated as specified in Section 4, shall not exceed the established maximum point values.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article of finished cloth shall be visually inspected for uniformity of shade, color, finish, and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. In addition, the contractor shall furnish a certificate of compliance for the cotton fiber identification, the carding of the cotton fibers, and the singeing and mercerization of the finished cloth.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

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4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4m) the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. All defects are defined in Section 1 of FED-STD-4, which are clearly noticeable at normal inspection distance, 3 feet (0.91m), shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only the slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "E" for slubs and "D" for knots shall be scored. No linear yard (increments of 1 yard (0.91m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four (4) points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 30.0 points. The lot shall be unacceptable if the individual points per 100 square yards ($83.6m^2$) of two or more individual rolls exceeds 45.0 points. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. If one or more rolls exceed the individual point value, the lot shall be rejected. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards } (83.6m^2)$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.6cm) in any dimension	= one point
For defects exceeding 3 inches (7.6cm) but not exceeding 6 inches (15.2cm) in any dimension	= two points
For defects exceeding 6 inches (15.2cm) but not exceeding 9 inches (22.9cm) in any dimension	= three points
For defects exceeding 9 inches (22.9cm) in any dimension	= four points

The following defects, when present, shall be scored 4 points for each yard (.91m) in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Poor dye penetration, mottled, streaky or cloudy
Overall uncleanness
Excessive neppiness
Shade defects - shaded side to side, side to center, or end to end
Characteristics of finish, hand, and shade not equal to the standard sample

4.4.2.2 Examination for length.

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4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.82m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade.

* 4.4.2.3.1 Examination for shade variation. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shade variation side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification, non-identification of preshrinking process, and non-conformance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends or placed on the wrong side, contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing, and marking comply with Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be four (4) continuous yards (3.64m), full width of the finished cloth. The lot size shall be expressed in units of one yard (0.91m). The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot Size (yards)	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table II - Test Methods

Characteristic	Requirement Paragraph	Test Method
Yarn ply:		
Warp	3.3.1	Visual 1/
Filling	3.3.1	Visual 1/
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4.1	Visual 1/
Sulfur content	3.5.1	2020
Colorfastness to:		
Light	3.5.3	5660
Crocking	3.5.3	5651
Perspiration	3.5.3	5680
Laundering (after 3 cycles)	3.5.3	5610 2/ 3/
pH	3.8	2811
Nonfibrous materials	3.6.1	2611
Dimensional stability	3.7	5550
Seam efficiency	3.9	5110 4/

1/ One determination per each sample unit and the results reported as "pass" or "fail".
2/ The specimens must be dried after each of the 3 laundering cycles.
3/ Only the stain of the cotton fiber shall be rated when evaluating the color transfer cloth.
4/ The needle size shall be 0.032 (± 0.001) inch [0.8 (± 0.025) mm] across the blade. The thread shall conform to V-T-276, Type IA1, top (needle) thread ticket No. 60, 3 ply, and bottom (looper) thread ticket No. 70, 3 ply.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the applicable requirements of PPP-P-1134.

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6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in a garment worn over the metallized radar frequency protection coverall.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Color of cloth if other than specified (see 3.5).
c. Width required (see 3.4.2).
d. Minimum length if other than specified (see 3.10).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition docuemtns regarding arrangements for examination, quantity, and testing and approval of the first article.

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids (see 3.1).

* 6.4 Fabric defect scales. Fabric Defect Replica Kits are available from Sears Roebuck and Company, Department 817 (Attn: BSC 23-29), Sears Tower, Chicago, IL 60684.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-82283B(NU)	Cloth, Twill, Cotton Lightweight

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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